

Multimodal Preference Elicitation and the Elimination of Dominated Decisions

SUBMISSION 42

TODO

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2 Model

Let our *decision task* be defined by $\mathcal{X}$ and $\mathcal{Y}$, which are respectively the space of possible decision inputs and outputs. A single input $X \in \mathcal{X}$ and a single output is $Y \in \mathcal{Y}$. Because not all outputs may valid for a given input X , we let $\mathcal{Y}(X) \subseteq \mathcal{Y}$ be the set of permissible outputs on input X . Let $\mathcal{R}$ be the set of all possible decision rules we might implement in our society, where a *decision rule* $R \in \mathcal{R}$ is a function $R : \mathcal{X} \rightarrow \Delta(\mathcal{Y})$ such that $R(X) \in \mathcal{Y}(X)$ for all X .

We now define a formal model capturing the preferences of interest: “how an individual wants the decision rule to work”. This model decouples two components of an individual’s revealed preferences: *static preferences* over decision rules, and *response behaviors*, which translate these static preferences to responses to questions — in ways that may depend on the learning environment. This decoupling is supported, for example, by the theory of *Inherent Preferences* [Simonson, 2008], which argues that while actual decisions are constructed on-the-fly (captured by our response behavior model $\mathcal{B}$), they are generated from stable underlying dispositions (captured by our static preferences model $\mathcal{M}$).

We introduce these two models $\mathcal{M}$ and $\mathcal{B}$, and subsequently how they are derived from and can be used to formally capture known phenomena from the behavioral sciences.

2.1 Static Preference Model $\mathcal{M}$

Conceptually, an individual’s static preference model $\mathcal{M}$ encompasses any component of their preferences that is immutable, i.e., not dependent on the elicitation environment. We assume that an individual’s static preferences is a collection of *frames*, where a frame is a preference relation over the rules in $\mathcal{R}$.

Definition 2.1. A frame is a preference relation.

Definition 2.2 (Frame). A frame is a preference relation $\succsim$ over $\mathcal{R}$; that is, between each pair of rules R and R' , there exists a preference relation in $\{\succ, \sim, \prec\}$. We assume this relation is transitive¹ and complete.

A frame represents a single valid perspective from which a person could evaluate decision rules. For example, to formalize some quotes from the introduction into frames:

Example

TODO

Example

TODO

The critical element of our model is that individuals can have multiple different frames. Formally, there is a fixed universe of frames $\mathcal{J}$. A static preference model then contains all these frames $\mathcal{M} = \{\succsim_j\}_{j \in [\mathcal{J}]}$. To capture the idea that certain frames may be more important than others to the individual, let frame have some inherent importance weighting $\omega_j \in \mathbb{R}_{\geq 0}$.

¹Here we adopt the standard definition: a preference relation over $\mathcal{R}$ is transitive iff for all $R, R', R'' \in \mathcal{R}$, if $R \succsim R'$ and $R' \succsim R''$, then $R \succsim R''$.

Global preferences over $\mathcal{R}$. Naturally, an individual's frames may specify very different preferences over rules, capturing the pluralistic conflict that is known to define human moral reasoning []. We will formalize these conflicts in how we define a *global preference relation* $\succsim_{\mathcal{M}}$ over $\mathcal{R}$, which captures how a person forms beliefs across these multiple frames. This relation is nonstandard because it can contain a fourth relation: $\bowtie$, representing incomparability, which arises naturally from inconsistency between frames.

In order to have a well-defined ordering over rules within a frame, we introduce a frame-specific utility function $u_j : \mathcal{R} \rightarrow \mathbb{R}$ such that $u_j(R) > u_j(R') \iff R \succ_j R'$ and $u_j(R) = u_j(R') \iff R \sim_j R'$. Within one frame, all preferences are transitive.

The frame level score for a comparison over R, R' is $U_j(R, R') = u_j(R) - u_j(R')$. Then, we take these primitives up and define the following scores:

Definition 2.3 (Global Rule Scores). We define two scores per pairwise comparison:

$$s^+(R, R') = \sum_{j: R \succ_j R'} \omega_j(u_j(R) - u_j(R')) \quad s^-(R, R') = \sum_{j: R' \succ_j R} \omega_j(u_j(R') - u_j(R))$$

This definition allows the “vote” per frame to not only be weighted by the weight of the frame but by the intensity preference over the pairwise rules. This decision, rather than simply considering the binary indicator of whether someone prefers the rule over the other, allows us to more flexibly represent existing methods.

We will now formalize how these scores lead to preference relations by defining two meaningful quantities from these scores. Let the *affective intensity* be the radius $r(R, R') = s^+(R, R') + s^-(R, R')$, which describes how strongly the person feels (i.e., how non-indifferent they are), and the *preference resolvability* $\theta(R, R') = \frac{s^+(R, R') - s^-(R, R')}{s^+(R, R') + s^-(R, R')}$, which quantifies how far the person is toward R dominating R' .

Then, we threshold both of these with thresholds τ and τ' , respectively:

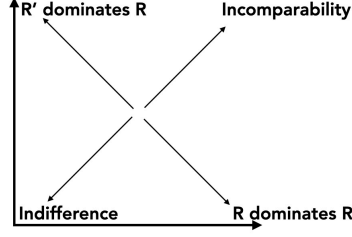
Definition 2.4 (Global Rule Preferences Induced by Utilities).

- $R \succ R'$ iff $r(R, R') > \tau$ and $\theta \geq \tau'$. This reflects the case where $s^+(R, R')$ is very large and $s^-(R, R')$ is very small; here, R tends to dominate R' on important frames, and thus R dominates R' .
- $R' \succ R$ iff $r(R, R') > \tau$ and $\theta \leq -\tau'$ (the converse of the above).
- $R \bowtie R'$ iff $r(R, R') > \tau$ and $\theta \in (-\tau', \tau')$. This reflects the case where $s^+(R, R')$ and $s^-(R, R')$ are very close together and very large, reflecting that highly important frames very much disagree, thus leading to incomparability.
- $R \sim R'$ iff $r(R, R') \leq \tau$. This reflects the case where $s^+(R, R')$ and $s^-(R, R')$ are both very small, so these two rules are almost always tied except on very unimportant frames. This means they should be tied.

Pictorially, $\succsim_{\mathcal{M}}$ varies as follows in space (where the axes are s^+, s^-):

We define preferences over pairwise comparisons only, which means that $\succsim_{\mathcal{M}}$ can be intransitive.

We now characterize when the utility-based representation of frames leads to intransitive global preferences over rules.



Recall that each frame j is represented by a normalized cardinal utility function $u_j : \mathcal{R} \rightarrow \mathbb{R}$, with importance weight $\omega_j \geq 0$. For any pair of rules R, R' , define the signed aggregate utility difference

$$m(R, R') := \sum_j \omega_j (u_j(R) - u_j(R')).$$

By Definition ??, this quantity satisfies

$$m(R, R') = s^+(R, R') - s^-(R, R'),$$

where s^+ and s^- are the positive and negative aggregate scores used to define the global pairwise preference relation.

Transitivity without thresholds. We begin by observing that intransitivity cannot arise if the thresholding geometry is removed.

Proposition 1 (Transitivity without thresholds). *If the decisiveness thresholds satisfy $\tau = 0$ and $\tau' = 0$, then the global preference relation $\succ_M$ is transitive. In particular,*

$$R \succ_M R' \iff m(R, R') > 0,$$

so $\succ_M$ coincides with the strict ordering induced by the scalar utility

$$U(R) := \sum_j \omega_j u_j(R).$$

PROOF. When $\tau = \tau' = 0$, Definition 1.3 reduces to the condition that $R \succ_M R'$ if and only if $s^+(R, R') \geq s^-(R, R')$ and the comparison is non-degenerate. Since $s^+ - s^- = m(R, R')$, this is equivalent to $m(R, R') > 0$, which in turn is equivalent to $U(R) > U(R')$. Thus $\succ_M$ is induced by a single scalar utility function and is therefore transitive. $\square$

Bailey: This is nice!

Proposition 1 shows that any global intransitivity in the model must arise from the introduction of nontrivial thresholds.

Intransitivity induced by thresholds. We now show that once $\tau > 0$, global intransitivity is possible even when all frame utilities are well-behaved.

THEOREM 1 (THRESHOLD-INDUCED INTRANSITIVITY). *Suppose $\tau > 0$. Then there exist frames $\{u_j\}$, weights $\{\omega_j\}$, and rules $A, B, C \in \mathcal{R}$ such that*

$$A \succ_M B, \quad B \succ_M C, \quad \text{but} \quad A \sim_M C.$$

In particular, the weak global preference relation $\preceq_M$ is not transitive.

PROOF. Consider two frames $j = 1, 2$ with weights

$$\omega_1 = 1, \quad \omega_2 = 1.5,$$

and thresholds $\tau = 1$ and $\tau' = 0.1$. Define frame utilities over three rules A, B, C as follows:

	A	B	C
u_1	0	1	-0.2
u_2	0	-1	-0.4

Comparison A vs. B . Frame 1 prefers B to A with intensity 1, while frame 2 prefers A to B with intensity 1. Thus

$$s^+(A, B) = 1.5, \quad s^-(A, B) = 1,$$

so $r = 2.5 > \tau$ and

$$\theta = \frac{1.5 - 1}{2.5} = 0.2 \geq \tau'.$$

Hence $A \succ_M B$.

Comparison B vs. C . Frame 1 prefers B to C with intensity 1.2, while frame 2 prefers C to B with intensity 0.6. Thus

$$s^+(B, C) = 1.2, \quad s^-(B, C) = 0.9,$$

so $r = 2.1 > \tau$ and

$$\theta = \frac{1.2 - 0.9}{2.1} \approx 0.143 \geq \tau'.$$

Hence $B \succ_M C$.

Comparison A vs. C . Both frames prefer A to C , but only weakly:

$$s^+(A, C) = 1 \cdot 0.2 + 1.5 \cdot 0.4 = 0.8, \quad s^-(A, C) = 0.$$

Thus $r = 0.8 \leq \tau$, and by Definition 1.3 we obtain $A \sim_M C$.

Combining the three comparisons yields $A \succ_M B$, $B \succ_M C$, but $A \sim_M C$, violating transitivity of $\succ_M$. $\square$

Interpretation. Theorem 1 shows that global intransitivity does not arise from inconsistencies within frames, but from the interaction between heterogeneous utility magnitudes and the decisiveness threshold τ . Even when utilities aggregate additively, low-intensity comparisons may be collapsed to indifference, producing failures of transitivity at the global level.

Bailey: Out of curiosity, can you also get intransitivity due to incomparability? I would assume so right, like $A \succ B, B \succ C, A \bowtie C$

Useful sets of decision rules. Finally, we partition the space of decision rules $\mathcal{R}$ into three sets based on $\succ_M$: *dominant rules* $\mathcal{R}^*$, *conflicted rules* $\mathcal{R}^?$, and *dominated rules* $\mathcal{R}^-$.

- **Dominated rules** are the clear losers (those strictly beaten by another rule):

$$\mathcal{R}^- := \{R \in \mathcal{R} : \exists R' \in \mathcal{R} \text{ s.t. } R' \succ_M R\}$$

Dominated rules are those that the agent's pluralistic system can consensually reject. Because at least one other rule R' is strictly preferred, R is never a rational choice for the agent. This includes rules trapped in preference cycles, which represent a failure of the system to establish a stable hierarchy rather than a lack of preference. **Is there some reason I'm missing that these should instead be conflicted?**

- **Dominant rules** are the undisputed winners among the undefeated:

$$\mathcal{R}^\star := \{R \in \mathcal{R} \setminus \mathcal{R}^- : \forall R' \in \mathcal{R} \setminus \mathcal{R}^-, (R \succ_{\mathcal{M}} R' \vee R \sim_{\mathcal{M}} R')\}$$

Dominant rules represent internal consensus: no matter which frame you look through, or how you aggregate them, these rules are at least as good as any viable alternative. When $\mathcal{R}^\star \neq \emptyset$, we call this *frame consistency* – i.e., the pluralistic frames can agree within the decision space.

- **Conflicted rules** are the disputed winners among the undefeated:

$$\mathcal{R}^? := \{R \in \mathcal{R} \setminus \mathcal{R}^- : \exists R' \in \mathcal{R} \setminus \mathcal{R}^- \text{ s.t. } R \succ_{\mathcal{M}} R' \}$$

Conflicted rules are those that are maximal (undefeated) but which represent a fundamental stalemate: the individual has conflicting frames which make it impossible to compare this rule against other undefeated rules. This set is the structural home of irreducible moral dilemmas, where pluralistic values are in a state of unresolvable tension.

Note that incomparabilities are necessary to produce conflicted rules; if there are no incomparabilities in $\succ_{\mathcal{M}}$ due to conflicts across frames, the first two sets partition the space of rules fully. Note also that these sets form a partition of $\mathcal{R}$: **TODO**.

Bailey: Need to formally address parametric models somewhere

2.1.1 Grounding $\mathcal{M}$ in behavioral science. The key departures of $\mathcal{M}$ from “standard” preference models in the AI alignment literature are (1) the inclusion of multiple, potentially conflicting frames; (2) the permission of indifference; (3) the permission of incomparability; (4) preferences being defined over rules instead of outcomes, and (5) the possibility of inherent intransitivity in preferences. We now ground these decisions in the behavioral sciences.

(1) Multiple, potentially conflicting frames. The inclusion of multiple, potentially conflicting frames is motivated by theories like *Moral Foundations Theory* [Haidt and Joseph, 2004] and *Value Pluralism* [Tetlock, 1986], which argue that humans’ moral reasoning cannot be captured by a single utility function; it is rather a collection of distinct “modules” (e.g., *fairness* or *care*, to take examples from Moral Foundations Theory) that are each internally consistent (corresponding to within-frame rankings with rational preferences) but may conflict with one another (corresponding to incomparabilities arising from disagreements across frames).

We explore some canonical examples of what we call “frames” documented in the behavioral science, and show how they are naturally formalized in our model.

CONSTRAINTS. In the introduction, we saw quotes like “*I think race should not be accounted for*” and “*I think mothers should always be prioritized*” and “*I don’t know if criminal record should be counted*”. These are examples of “constraints” (or non-constraints?) – rules that technically constrain the behavior of the decision rule across inputs. Constraints of this nature are well-documented in the XX literature... **TODO**. We show how each constraint of the form in the quote above produces a frame:

Example: “*I think race should not be a factor*” we can interpret this as saying that if our query X includes a dilemma between two people are otherwise identical but have different races, they should be identically the same by R . At minimum defines a behavior of R on all queries with this property, and says nothing about queries without this property – that is, unless a more structured parametric model is imposed on $\mathcal{R}$. E.g., if $\mathcal{R}$ is the class of all linear functions of the features, then $\beta_{\text{race}=} = 0$ for all values of race $\cdot$.

$\mathcal{R}$; as we show in Section XX, one can *derive* preferences over outcomes from preferences over rules. This modeling relates to economic ideas on *Meta-Preferences* [Grofman and Uhlaner, 1985, Sen, 1977], which suggests people hold stable second-order preferences (preferences about how preferences should be formed) that remain static even when first-order choices vary.

Should better understand this class of theories and put this somewhere: One entire class of theories that could contradict our model – depending on how it is interpreted – is that global preferences (in our case, over rules) are *constructed* from knee-jerk reactions. If one interprets knee-jerk reactions as *query responses*, then this directly contradicts our model and aligns more closely with existing models. However, this interpretation is not really borne out in the data – it seems that people have articulable preferences over rule behavior, that often seem much more decisive than their query responses. One could alternatively treat these “knee-jerk” reactions more generally, e.g., what initially seems salient based on the query. This will be consistent with our behavior response model in the next section.

(5) The possibility of inherent intransitivity. The conceptual interpretation of ?? is that in our model, intransitive preferences are not necessarily a result of behavioral deviations; they can also be part of someone’s true preferences, arising due to multiple conflicting beliefs (frames). This aligns with some arguments in the behavioral sciences and contradicts some others: [Cover literature here](#)

2.2 Response Behavior Model $\mathcal{B}$

When we aim to learn “how an individual wants the rule to work, we will ask them questions called *queries*, possibly in one or more formats. A **query format** is its own decision task, with possible inputs $\mathcal{Q}$ and possible outputs $\mathcal{W}$, with elements $Q \in \mathcal{Q}$ being a single query and $W \in \mathcal{W}$ being a single response. As before, we say that the valid outputs associated with a query Q is $\mathcal{W}(Q) \subseteq \mathcal{W}$. For example, in the supervised learning approach, people are queried on instances of the decision problem itself, so $\mathcal{Q} \subseteq \mathcal{X}$ and $\mathcal{W} \subseteq \mathcal{Y}$, but this is not always the case, and it certainly does not need to be; we will discuss the relationship between the query space $\mathcal{Q}$, $\mathcal{W}$ and the decision space $\mathcal{X}$, $\mathcal{Y}$ in Section XX.

Here, we now formalize how the static elements of people’s models $\mathcal{M}$ translate to query responses, which we allow to depend on two things: the *query format* defined above and the broader *elicitation context* defined below. We leave the precise mapping very general because, as illustrated by section XX, there are an enormous number of theories that could instantiate it. At a high level, we should think of the response behavior model as capturing how preferences are constructed based on $\mathcal{M}$, in response to some query Q , and in some *elicitation context*.

The **elicitation context** $\mathcal{C}$ encompasses anything else around an individual query modeled as relevant to how a person responds to the query. It could consist of all the queries asked before, interpersonal influence in a deliberative context, or any other element of the scenario that is influencing how to render a decision when given a query Q . We leave $\mathcal{C}$ totally general for now, with the recognition that it could be formally modeled when relevant; it exists here mostly for conceptual accounting, and it will be useful in relating our model to the literature.

Given a decision space a static preference model $\mathcal{M}$ (encompassing the decision setting $\mathcal{X}$, $\mathcal{Y}$, $\mathcal{R}$, a query space $\mathcal{Q}$, $\mathcal{W}$, and a context $\mathcal{C}$, a *response behavior model* is a (potentially randomized) mapping $\mathcal{B}_{\mathcal{M}, \mathcal{C}} : \mathcal{Q} \rightarrow \Delta(\mathcal{W})$, where $\Delta(\cdot)$ represents the set of all distributions over a set. In other words, the response behavior model defines how the person translates their preference model to a query response under a given context.

The *Pareto-optimal set* (Pareto frontier) of $\mathcal{H}$ is

$$\text{Pareto}(\mathcal{H}) := \left\{ R \in \mathcal{H} : \nexists R' \in \mathcal{H} \text{ that Pareto-dominates } R \right\}.$$

We view the learning goal as (a) foremost, outputting the entire set $\text{Pareto}(\mathcal{H})$ (or at least providing a polynomial time oracle for querying whether a certain rule is in it); (b) outputting some rule in $\text{Pareto}(\mathcal{H})$.

Bailey: Does this always exist? Or what always exists

ε -Pareto dominance and ε -Pareto optimality. When exact learning is impossible, we want to output a rule that is *approximately* Pareto-optimal.

Fix $\varepsilon \geq 0$. We say that $R' \in \mathcal{H}$ ε -Pareto-dominates $R \in \mathcal{H}$ if

$$u_j(R') \geq u_j(R) + \varepsilon \quad \forall j \in [m].$$

A rule $R \in \mathcal{H}$ is ε -Pareto-optimal (equivalently, ε -Pareto-undominated) if there does not exist any $R' \in \mathcal{H}$ that ε -Pareto-dominates it, i.e.,

$$R \in \text{Pareto}_\varepsilon(\mathcal{H}) \iff \nexists R' \in \mathcal{H} \text{ such that } u_j(R') \geq u_j(R) + \varepsilon \quad \forall j \in [m].$$

Remark (alternative ε notion). Some authors use a weaker ε -dominance notion: R' dominates R if $u_j(R') \geq u_j(R)$ for all j and $u_{j^*}(R') \geq u_{j^*}(R) + \varepsilon$ for some j^* . The definition above is the *uniform-margin* version, requiring an ε improvement for every frame.

3 Pairwise Comparison Queries under Pluralistic Moral Frames

A classical approach is to show people *examples* of how a rule could behave on inputs and ask them to pin down the behavior of the rule. A very specific way of doing this is *pairwise comparison queries* limits output-space comparisons to two options, and are as follows:

Definition 3.1 (Pairwise Comparison Query). A pairwise comparison query is defined as

$$Q := \{(X, Y, Y') | X \in \mathcal{X}, Y, Y' \in \mathcal{Y}(X)\}$$

where $\mathcal{X}$ is the decision input space and $\mathcal{Y}(X)$ is the set of permissible outputs on input X .

Despite their popularity, there is general suspicion that pairwise comparisons are failing to capture true beliefs. However, this is hard to argue formally in existing preference models, in which pairwise comparison-based learning actually *does* recover the truth. We now explore this critique in our model, where people can have pluralistic moral frames.

First, we formalize a model of how our preference model over rules translates to a preference model over pairwise outcomes (). Then, we show that the skepticism about pairwise comparisons is warranted: we give realistic conditions under which pairwise comparisons lead to crucial identification failures (), and show that even under sufficient conditions for identification, learning via pairwise comparisons can fail catastrophically under standard assumptions ().

3.1 Model: Preferences over Pairwise Comparisons

To move from global preferences over rules to local choices in specific cases, we must specify how an agent evaluates outcomes as evidence about the rule governing behavior. We adopt a rule-first perspective: **rules are the primary objects of evaluation**, and observed outcomes provide partial information about how well a rule aligns with an individual's values. Pairwise comparison queries are therefore interpreted as asking which outcome, if observed at a given input, would be more indicative of a desirable underlying rule.

Within-frame preferences. First, we define how a *single frame* implies preferences over a pairwise comparison. We treat each individual frame as a “version of the person”, where they are using their fundamental preferences over rules to define projections:

$$s_j^+(X, Y, Y') = \sum_{R: R[X \rightarrow Y] \succ_j R[X \rightarrow Y']} u_j(R[X \rightarrow Y]) - u_j(R[X \rightarrow Y'])$$

$$s_j^-(X, Y, Y') = \sum_{R: R[X \rightarrow Y] \prec_j R[X \rightarrow Y']} u_j(R[X \rightarrow Y']) - u_j(R[X \rightarrow Y])$$

As usual, our preference relation is defined like this:

- If $s_j^+ \gg s_j^-$ and sufficiently large, $Y \succ Y'$: $s^+ + s^- \geq \tau$ and $\frac{s^+ - s^-}{s^+ + s^-} \geq \tau'$.
- $Y \sim_j Y'$ if $s^+ + s^- < \tau$
- $Y \bowtie_j Y'$ if $s^+ + s^- \geq \tau$ and $\frac{s^+ - s^-}{s^+ + s^-} \in (-\tau', \tau')$

Definition 3.2 (Separability). A frame j is *separable at input X* if for all $Y, Y' \in \mathcal{Y}(X)$ and all $R, R' \in \mathcal{R}$

$$R[X \mapsto Y] \succ_j R[X \mapsto Y'] \iff R'[X \mapsto Y] \succ_j R'[X \mapsto Y'].$$

We are assuming that all projections exist in the rulespace.

Cross-frame preferences. Now, we define how within-frame preferences imply cross-frame preferences. The way we define this relative to previous definitions reflects the following narrative of reasoning: A person evaluates their preferences within each frame separately first, as each frame is an individual internal “wolf”. Then, these within frame preferences are aggregated.

Definition 3.3 (Global Outcome Scores). Fix a query $q = (X, Y, Y')$. Let

$$U_j(Y) = \sum_R u_j(R[X \rightarrow Y]) \quad U_j(Y') = \sum_R u_j(R[X \rightarrow Y'])$$

Then define the scores of different answers as

$$s^+(Y, Y') := \sum_{j: Y \succ_j Y'} \omega_j(U_j(Y) - U_j(Y')) \quad s^-(Y, Y') := \sum_{j: Y \prec_j Y'} \omega_j(U_j(Y') - U_j(Y))$$

Definition 3.4 (Query-Model Outcome Preferences). We again define the quantities we will threshold:

$$r = s^+ + s^-, \quad \theta = \frac{s^+ - s^-}{s^+ + s^-}$$

Now, we can define analogous conditions to global preferences over rules, now over Y, Y' with respect to a single query X :

- $Y \succ_{\mathcal{M}, X} Y' \iff r \geq \tau \text{ and } \theta \geq \tau'$
- $Y' \succ_{\mathcal{M}, X} Y \iff r \geq \tau \text{ and } \theta \leq -\tau'$
- $Y \bowtie_{\mathcal{M}, X} Y' \iff r \geq \tau \text{ and } \theta \in (-\tau', \tau')$
- $Y \sim_{\mathcal{M}, X} Y' \iff r < \tau$

An outcome at a given input is evaluated not for its immediate consequences alone, but for what it reveals about the underlying decision rule; frames prefer outcomes that are more indicative of rules aligned with their values. Each frame j is assumed to have a (possibly implicit) behavioral model describing how a rule R gives rise to outcomes, represented by a likelihood $P(Y|X, R)$. This likelihood need not be accurate or known to the learner; it formalizes how individuals interpret outcomes as evidence about rule behavior.

Definition 3.5 (Active frames). For a given query $q = X, Y, Y'$, a frame j is *active* iff $Y \succ_j Y'$ or $Y \prec_j Y'$. We denote the set of active frames on a query as $\mathcal{J}(q)$.

Any frame can be written as an *objective* (where rules are then evaluated on this objective and there is some structured ordering of rules depending on how they perform in this objective), or a *constraint* (where rules are partitioned into two equivalence sets – those that always satisfy the constraint or don't).

3.2 Outcome-Based Frames

Recall that each frame j is a complete, transitive preference relation $\succ_j$ over rules $\mathcal{R}$, represented by a cardinal utility $u_j : \mathcal{R} \rightarrow \mathbb{R}$ via $u_j(R) > u_j(R') \iff R \succ_j R'$. Also recall the projection notation $R[X \mapsto Y]$ and the definition of separability at an input X . Throughout, we assume (as in the paper) that all projections exist in $\mathcal{R}$.

Outcome-based frames (how the rule distributes benefits and harms). We let $W_i(X, Y)$ be i 's welfare associated with outcome Y in query X .

Definition 3.6 (Outcome-based). A frame j is *outcome-based frame* if, for a fixed distribution over inputs $\mathcal{D}$, a *local aggregator* $\phi_j : \mathcal{X} \rightarrow \mathcal{Y} \rightarrow \mathbb{R}$, and an *outer aggregator* $F_j : \mathbb{R} \rightarrow \mathbb{R}$ weakly increasing, it holds that

$$u_j(R) = F_j(\mathbb{E}_{X \sim \mathcal{D}}[\phi_j(X, R(X))]). \quad (1)$$

(If rules are randomized, interpret $\phi_j(X, R(X))$ as $\mathbb{E}_{Y \sim R(X)}[\phi_j(X, Y)]$.)

Implicitly, ϕ should be a function of the welfares — unless there is some other outcome benefit?

Remark. The special case $F_j(t) = t$ yields the familiar “expected additive score” form $u_j(R) = \mathbb{E}[\phi_j(X, R(X))]$, which includes utilitarian-style objectives and many weighted variants. In allocation problems, ϕ_j can itself be built from $W_i(X, Y)$ (e.g., $\phi_j(X, Y) = \sum_{i \in U} W_i(X, Y)$).

COROLLARY 2 (OUTCOME-METRIC FRAMES: A CLEAN SUFFICIENT (AND ESSENTIALLY NECESSARY) CONDITION). *Consider an outcome-metric frame of the form $u_j(R) = F_j(\mathbb{E}_{X \sim \mathcal{D}}[\phi_j(X, R(X))])$ as in equation (1). If F_j is (strictly) increasing, then the frame is separable at every input X . In the finite- X full-product setting, the converse holds: separability at every X implies an additive form $u_j(R) = c + \sum_X v_{j,X}(R(X))$, i.e. u_j is an (increasing transform of an) expected pointwise score.*

PROOF SKETCH. For the forward direction, fix X and compare $R[X \mapsto Y]$ vs. $R[X \mapsto Y']$: the only term in $\mathbb{E}[\phi_j(\cdot, R(\cdot))]$ that changes is the contribution at X (scaled by D), so the sign of the utility difference depends only on $\phi_j(X, Y) - \phi_j(X, Y')$, not on R . The converse in the finite full-product case follows from Theorem ?? by absorbing constants/scales into $v_{j,X}$. $\square$

Examples. There are many types of frames of this sort.

- **Utilitarianism:** this maximizes the total benefit. This means that for fixed $\mathcal{D}$, $R \succ_j R'$ iff

$$\mathbb{E}_{X \sim \mathcal{D}}[\sum_{i \in U} \mathcal{W}_i(X, R(X))] \geq \mathbb{E}_{X \sim \mathcal{D}}[\sum_{i \in U} \mathcal{W}_i(X, R'(X))]$$

Here, the inner aggregator ϕ is the sum of all utilities, and the outer aggregator F_j is the identity function.

Example: in the trolley problem, we could define everyone’s welfare function as 0 if they get hit, 1 if they don’t get hit. This would mean that the dominant rule for *any* input distribution is the one that saves the most people in every instance.

- **Egalitarianism:** Means that no one should be treated too poorly by the mechanism. This means that for fixed $\mathcal{D}$, $R \succ_j R'$ iff

$$\min_{i \in U} \mathbb{E}_{X \sim \mathcal{D}}[\mathcal{W}_i(X, R(X)) | i \in X] \geq \min_{i \in U} \mathbb{E}_{X \sim \mathcal{D}}[\mathcal{W}_i(X, R'(X)) | i \in X]$$

Here, the inner aggregator ϕ is the welfare of i conditional on showing up, and the outer aggregator is the min over all people.

Example: In the kidney exchange problem suppose that everyone has the same welfare function: $\mathcal{W}_i(X, R(X)) = 1$ if they get the kidney and $\mathcal{W}_i(X, R(X)) = 0$ otherwise. Then, this tries to make sure that whenever i shows up, conditional on showing up they have the same expected welfare. A dominant rule in this frame is any rule such that maximizes

$$\max_{R \in \mathcal{R}} \min_{i \in U} \mathbb{E}_{X \sim \mathcal{D}}[\mathcal{W}_i(X, R(X)) | i \in X]$$

I don't think there is a single dominant rule for this because the objective is kind of degenerate, and an ordering of rules *can only be defined for a fixed inputs distribution*.

- **Group proportionality:** Fix some group G , and let it be the case that either $i \in G$ or $i \notin G$. We may want to say that of the total benefit received by anyone, we want group G 's members to receive exactly 30%. This could mean that $R \succ_j R'$ iff

$$\mathbb{E}_{X \sim \mathcal{D}} \left[\frac{\sum_{i \in G} \mathcal{W}_i(X, R(X))}{\sum_{i \in U} \mathcal{W}_i(X, R(X))} \right] = 0.3 \quad \wedge \quad \mathbb{E}_{X \sim \mathcal{D}} \left[\frac{\sum_{i \in G} \mathcal{W}_i(X, R'(X))}{\sum_{i \in U} \mathcal{W}_i(X, R'(X))} \right] \neq 0.3$$

or you could measure some distance from 0.3 and order rules that way. Either way, there are many dominant rules in this frame, all of which achieve the 0.3 exactly.

- **Membership to group G should not affect benefits.** This means that if two people differ only on their membership to group G , then they should be treated identically. This can be seen as a constraint on rules, i.e., R satisfies this constraint iff for all i, i' who are identical group membership-wise except $i \in G$ and $i \notin G$,

$$\mathbb{E}_{X \sim \mathcal{D}} [\mathcal{W}_i(X, R(X))] = \mathbb{E}_{X \sim \mathcal{D}} [\mathcal{W}_{i'}(X, R(X))]$$

Does this go in this category? I think so, you can make F_j the indicator of whether the difference = 0.

- **People in group G deserve more benefits.** Or we might want a rule that on average benefits that group more: $R \succ_j R'$ iff

$$\mathbb{E}_{X \sim \mathcal{D}} \left[\sum_{i \in G} \mathcal{W}_i(X, R(X)) \right] \geq \mathbb{E}_{X \sim \mathcal{D}} \left[\sum_{i \in G} \mathcal{W}_i(X, R'(X)) \right]$$

Proposition 2. Maximin as defined above is inseparable

PROOF. A frame is *inseparable* on query X iff there exists R, R' such that

$$R[X \rightarrow Y] \succ_j R[X \rightarrow Y'] \quad \text{and} \quad R'[X \rightarrow Y] \prec_j R'[X \rightarrow Y'].$$

Fix a $\mathcal{D}$, which induces a preference relation over rules R such that

$$\min_Y \Pr_X[R(X) = Y] > \min_Y \Pr_X[R'(X) = Y] \iff R \succ R'.$$

There exists a distribution where separability is violated. Let's suppose there are two possible inputs X that are equally likely. There are two possible food recipients in the system, A and B , and each input corresponds to a different donation and we have to decide whether to allocate it to A or B .

Then, there are four possible rules:

- (1) $X_1, X_2 \rightarrow A$
- (2) $X_1, X_2 \rightarrow B$
- (3) $X_1 \rightarrow A, X_2 \rightarrow B$
- (4) $X_1 \rightarrow B, X_2 \rightarrow A$

Rules 3 and 4 are preferred to 1 and 2, because they maximize the minimum welfare, distributing the goods equally. However, we can find two disagreeing projections, because each replacement *can* be wrong depending on what goes on elsewhere.

Inseparability at X_2 :

$$R_4[X_2 \rightarrow A] = R_4 \succ R_2 = R_4[X_2 \rightarrow B] \quad \wedge \quad R_3[X_2 \rightarrow A] = R_1 \prec R_3 = R_3[X_2 \rightarrow B]$$

Inseparability at X_1 :

$$R_3[X_1 \rightarrow A] = R_3 \succ R_2 = R_3[X_1 \rightarrow B] \quad \wedge \quad R_4[X_1 \rightarrow A] = R_1 \prec R_4 = R_4[X_1 \rightarrow B]$$

Is inseparability here specific to the distribution over inputs? To think about this, let's pretend that in the above example, X_1 appears with probability ϵ and X_2 appears with probability $1 - \epsilon$. *the example still survives*. So basically we need separability over all X 's that actually can appear in the wild, otherwise an example like the above can happen. $\square$

3.3 Axiomatic frames (how the rule treats different inputs)

These frames govern how a rule should act on either one input or a group of inputs. These are most naturally formulated as *constraints*, but can be relaxed (i.e., frequency of satisfaction).

Definition 3.7 (Axiomatic (relational) frame). Fix a distribution $\mathcal{D}$ over inputs. A frame j is axiomatic if it is specified by:

- (1) an arity $k_j \geq 1$;
- (2) a distribution κ_j over tuples $(X_1, \dots, X_{k_j}) \in \mathcal{X}^{k_j}$ whose marginals are $\mathcal{D}$ (e.g. a matched-pairs coupling);
- (3) a local axiom predicate $A_j : \mathcal{X}^{k_j} \times \mathcal{Y}^{k_j} \rightarrow \{0, 1\}$;

and it evaluates rules by the (expected) axiom satisfaction rate

$$\text{viol}_j(R) := \mathbb{E}_{(X_1, \dots, X_{k_j}) \sim \kappa_j} \left[1 - A_j((X_1, \dots, X_{k_j}), (R(X_1), \dots, R(X_{k_j}))) \right], \quad (2)$$

through some weakly decreasing outer map $G_j : [0, 1] \rightarrow \mathbb{R}$:

$$u_j(R) = G_j(\text{viol}_j(R)). \quad (3)$$

The hard-constraint special case is $G_j(t) = 0$ if $t = 0$ and $G_j(t) = -\infty$ if $t > 0$, equivalently the constraint set $C_j := \{R : \text{viol}_j(R) = 0\}$.

Remark 1. Outcome-based frames factor through one-point information $(X, R(X))$ and then aggregate. Axiomatic frames factor through k -point (typically two-point) relations among outputs $(R(X_1), \dots, R(X_{k_j}))$ on coupled inputs $(X_1, \dots, X_{k_j})$. This higher-arity dependence is what makes them “axiomatic” in a formal sense.

When is an axiomatic frame separable? The punchline is: an axiomatic frame is separable iff it is actually unary/pointwise (i.e., its axioms do not impose cross-input relations).

To state this cleanly, it is helpful to isolate the “pointwise factorization” condition.

Definition 3.8 (Pointwise factorization). An axiomatic frame $(k_j, \kappa_j, A_j, G_j)$ factorizes pointwise if $k_j = 1$ and there exists a predicate $\tilde{A}_j : \mathcal{X} \times \mathcal{Y} \rightarrow \{0, 1\}$ such that

$$A_j((X_1), (Y_1)) \equiv \tilde{A}_j(X_1, Y_1).$$

Equivalently, the hard-constraint set C_j can be written as

$$C_j = \left\{ R \in \mathcal{R} : \forall X \in \mathcal{X}, R(X) \in \mathcal{A}_j(X) \right\} \quad \text{for some sets } \mathcal{A}_j(X) \subseteq \mathcal{Y}(X). \quad (4)$$

Proposition 3 (Axiomatic separability: characterization in the full-product finite case). Assume $\mathcal{X}$ is finite and $\mathcal{R} = \prod_{X \in \mathcal{X}} \mathcal{Y}(X)$ (every pointwise combination is feasible). Fix an axiomatic frame as in Definition 3.7 with G_j strictly decreasing. Then the following are equivalent:

- (1) The frame is separable at every input X (in the sense of Definition 3.2 of your paper).
- (2) The frame factorizes pointwise (Definition 3.8), i.e. it imposes only unary constraints.

In particular, any nontrivial $k_j \geq 2$ axiom (sensitive to the joint behavior on distinct inputs) is inseparable.

PROOF SKETCH. $(2 \Rightarrow 1)$ If constraints factorize as equation (4), then the acceptability/penalty contribution from input X depends only on $R(X)$; hence the preference between $R[X \mapsto Y]$ and $R[X \mapsto Y']$ does not depend on what R does elsewhere.

$(1 \Rightarrow 2)$ Contraposition. Suppose the axiom is genuinely relational: there exist a tuple $(X_1, \dots, X_{k_j})$ in the support of κ_j with at least two distinct coordinates and outcomes $(y_1, \dots, y_{k_j}), (y'_1, \dots, y'_{k_j})$ that differ only at coordinate ℓ such that

$$A_j((X_1, \dots, X_{k_j}), (y_1, \dots, y_{k_j})) \neq A_j((X_1, \dots, X_{k_j}), (y'_1, \dots, y'_{k_j})).$$

Using the full-product rule space, build two ambient rules R, R' that agree everywhere except at some other coordinate $m \neq \ell$ so that the axiom satisfaction at the tuple flips. Then the preference between the two projections at X_ℓ depends on whether the ambient rule sets $R(X_m)$ or $R'(X_m)$, violating separability. $\square$

Remark 2 (Interpretation). This proposition is exactly the “formal reason” that counterfactual invariance constraints like

$$R(X) = R(X') \quad \text{whenever } (X, X') \text{ form a matched pair under } \kappa_j$$

are inseparable: they are genuinely two-point statements. By contrast, pointwise admissibility constraints (“at each X , outcomes must lie in an allowed set”) are axiomatic but separable.

Examples:

- **Ignoring group membership.** *I don't think (membership to group G) should matter.*

One weird thing though is that exact equality may be impossible to satisfy because i, i' may show up in very different kinds of situations in $\mathcal{D}$. There is a weaker version of this, which is that for any two inputs X, X' that are identical except that $i \in G$ is replaced with someone else who is identical except $i' \notin G$, it must be that

$$R(X) = R(X')$$

Formally, Let $k_j = 2$ and let κ_j couple each X with its counterfactual swap X' (same non-protected features, group label flipped). Let

$$A_j((X, X'), (Y, Y')) := \mathbf{1}\{Y = Y'\}.$$

Then $\text{viol}_j(R)$ is the probability (under matched pairs) that R changes its decision when only group membership changes.

- **Prioritizing members of a certain group.** *I think that members of group G should always be prioritized.* This could mean a lot of different things, either at the query level or the distribution level.

- Or we could be saying that on two inputs where someone is shifted from outside that group to in that group and nothing else changes, they should never be harmed. R satisfies this constraint iff for any such two X, X' (where the person i is in G in X and not in X' but all else is held fixed):

$$\mathcal{W}_i(X, R(X)) \geq \mathcal{W}_i(X', R(X')) \quad \forall X$$

One could also define the randomized version of this, where X, X' are drawn randomly and then we are evaluating the probability of R satisfying this constraint, and the best R satisfies it.

Formally, Let κ_j couple (X, X') where X' is obtained from X by changing one individual's group membership $i \notin G$ to $i \in G$ holding all other features fixed. You can encode “never harm by moving into G ” either as:

$$A_j((X, X'), (Y, Y')) := \mathbf{1}\{\mathcal{W}_i(X, Y) \geq \mathcal{W}_i(X', Y')\},$$

or, if you prefer to avoid welfare, by an outcome-level dominance relation (if outcomes are directly comparable). This is axiomatic even though it mentions welfare: the formal reason is that it compares two inputs.

- **I think all members of two groups, G_1, G_2 should be treated equally.** This can be interpreted many different ways.

- For example, it could mean that if two queries are identical except the members of group G_1 are replaced with otherwise identical members of G_2 , the decision shouldn't change. Formally, R satisfies this constraint iff for all X, X' as described above,

$$R(X) = R(X').$$

- It could be that this is about total distribution of benefits being equal in expectation across the true groups, conditional on their members showing up

3.4 Takeaways

There are two kinds of frames — frames that look at some function of the distribution of the expected benefits over some input distribution $\mathcal{D}$, and frames that describe constraints on individual groups of inputs.

Both types can be separable / inseparable.

4 Our Learning Algorithm: Linear Frames

We will now instantiate our model under assumptions that closely reflect — but still generalize — common preference learning assumptions made in existing AI alignment work. This version of our model is the analog of learning over a class of linear models over a d -dimensional feature space, as done in [10, 11].

Suppose that each input, output combination exists within some latent d -dimensional feature space. Formally, let there be some function $\phi : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}^d$ such that $\phi(X, Y)$ represents the feature vector of X, Y .

Definition 4.1 (Linear rules). For any $\theta \in \mathbb{R}^d$, a rule R_θ is *linear* in θ if it acts according to linear scoring:

$$R_\theta(X) \in \arg \max_{Y \in \mathcal{Y}(X)} \theta \cdot \phi(X, Y)$$

Definition 4.2 (Linear frames). A frame j is *linear* with respect to $\Theta \subseteq \mathbb{R}^d$ iff there exists $v_j \in \mathbb{R}^d$ and $b_j \in \mathbb{R}^d$ such that for all $\theta, \theta' \in \Theta$,

$$R_\theta \succ_j R_{\theta'} \iff \langle v_j, \theta \rangle + b_j \geq \langle v_j, \theta' \rangle + b_j \iff \langle v_j, \theta \rangle \geq \langle v_j, \theta' \rangle$$

Let $V = (v_1^T, \dots, v_m^T)$, so it is $d \times m$ matrix.

Extension to projective closure of preferences. Above, we define $u_j(R_\theta) = \langle v_j, \theta \rangle$, which gives us preference over all *linear* rules. To use our projection definitions, we need to actually have a defined preference model over *all* rules, not just all linear rules. We define an extension of the above which gives the same utilities for linear rules, but also which specifies utilities for all other rules as well.

Fix any scaling $\lambda : \mathcal{X} \rightarrow (0, \infty)$ (e.g. $\lambda(X) \equiv 1$; or $\lambda(X) = \Pr_{X' \sim \mathcal{D}}[X' = X]$ if $\mathcal{X}$ is finite). For a base linear rule R_θ and a finite set of edits

$$E = \{(X_1, Y_1), \dots, (X_k, Y_k)\} \subseteq \{(X, Y) : X \in \mathcal{X}, Y \in \mathcal{Y}(X)\},$$

assume the edited inputs are distinct (i.e. $X_r \neq X_s$ for $r \neq s$). Define the *edited rule* $R_{\theta, E} \in \mathcal{R}$ by

$$R_{\theta, E}(X) := \begin{cases} Y_i, & \text{if } X = X_i \text{ for some } (X_i, Y_i) \in E, \\ R_\theta(X), & \text{otherwise.} \end{cases}$$

We define the **extended frame utility** of $R_{\theta, E}$ by

$$u_j(R_{\theta, E}) := u_j(R_\theta) + \sum_{(X_i, Y_i) \in E} \lambda(X_i) \left(\langle v_j, \phi(X_i, Y_i) \rangle - \langle v_j, \phi(X_i, R_\theta(X_i)) \rangle \right). \quad (5)$$

Consistency and reduction to the single-edit case. If $E = \emptyset$, then $R_{\theta, E} = R_\theta$ and equation (5) gives $u_j(R_{\theta, \emptyset}) = u_j(R_\theta) = \langle v_j, \theta \rangle$.

Path-independence (order of edits does not matter). Because equation (5) is additive over the set E (with distinct edited inputs), $u_j(R_{\theta, E})$ depends only on the set of edits, not the order in which they are applied.

Cardinal replacement gaps are well-defined. Fix an edited rule $R_{\theta,E}$ and an input $X \notin \{X_i : (X_i, Y_i) \in E\}$. For any $Y, Y' \in \mathcal{Y}(X)$,

$$u_j(R_{\theta,E \cup \{(X,Y)\}}) - u_j(R_{\theta,E \cup \{(X,Y')\}}) = \lambda(X) \langle v_j, \phi(X, Y) - \phi(X, Y') \rangle,$$

which is independent of the ambient edited rule $R_{\theta,E}$.

Implication of strong separability. Under this construction of preferences, cardinal separability holds:

Definition 4.3 (cardinal separability). A frame j is *cardinally separable at input X* (over a rulespace $\mathcal{S} \subseteq \mathcal{R}$) if for all $Y, Y' \in \mathcal{Y}(X)$ and all $R, R' \in \mathcal{S}$,

$$u_j(R[X \mapsto Y]) - u_j(R[X \mapsto Y']) = u_j(R'[X \mapsto Y]) - u_j(R'[X \mapsto Y']).$$

We say j is *cardinally separable on $\mathcal{S}$* if it is strongly separable at every $X \in \mathcal{X}$.

Proposition 4 (Cardinal separability on the projective closure under the finite-edits utility). Fix $\lambda : \mathcal{X} \rightarrow (0, \infty)$ and define utilities on the projective closure via equation (5). Let

$$\mathcal{S} := \{R_{\theta,E} : \theta \in \Theta, E \text{ is a finite set of edits with distinct inputs}\}$$

be the resulting projective closure of $\mathcal{H}$. Then for every frame $j \in [m]$ and every input $X \in \mathcal{X}$,

$$u_j(R_{\theta,E}[X \mapsto Y]) - u_j(R_{\theta,E}[X \mapsto Y']) = \lambda(X) \langle v_j, \phi(X, Y) - \phi(X, Y') \rangle \quad (6)$$

whenever X is not already edited in E . In particular, the replacement gap is independent of the ambient rule in $\mathcal{S}$, and hence frame j is cardinally separable at X over $\mathcal{S}$.

PROOF. Fix θ , an edit set E with distinct edited inputs, and an unedited input $X \notin \{X_i : (X_i, Y_i) \in E\}$. By definition, $R_{\theta,E}[X \mapsto Y] = R_{\theta,E \cup \{(X,Y)\}}$ and likewise for Y' . Applying equation (5) to these two edited sets yields

$$\begin{aligned} u_j(R_{\theta,E}[X \mapsto Y]) &= u_j(R_\theta) + \sum_{(X_i, Y_i) \in E} \lambda(X_i) \left(\langle v_j, \phi(X_i, Y_i) \rangle - \langle v_j, \phi(X_i, R_\theta(X_i)) \rangle \right) \\ &\quad + \lambda(X) \left(\langle v_j, \phi(X, Y) \rangle - \langle v_j, \phi(X, R_\theta(X)) \rangle \right), \end{aligned}$$

and the same expression with Y replaced by Y' for $u_j(R_{\theta,E}[X \mapsto Y'])$. Subtracting cancels all terms except the last, giving

$$u_j(R_{\theta,E}[X \mapsto Y]) - u_j(R_{\theta,E}[X \mapsto Y']) = \lambda(X) \left(\langle v_j, \phi(X, Y) \rangle - \langle v_j, \phi(X, Y') \rangle \right),$$

which is equation (6). The right-hand side depends only on (X, Y, Y') , not on (θ, E) , so the replacement gap is independent of the ambient rule, establishing strong separability. $\square$

Hypothesis class. We will define our hypothesis class as the set of all linear rules that have a linear relationship with the frames via the v_j and the norm be $p = 2$.

$$\Theta = \left\{ \frac{V\omega}{\|V\omega\|_p} \mid \omega \in \mathbb{R}^m \right\}.$$

In our learning algorithm, V will be known to the learner at the time the hypothesis class needs to be defined via a “pre-learning” period. **Conceptually, this corresponds to ...**

Bailey: Knowing V_j — what does this conceptually mean?

We now show why our preference model and hypothesis class, even under these restrictions, generalize the popular approach of using pairwise comparisons with the naive decoder. First, we define the linear hypothesis class used in, e.g., [10], as $\mathcal{B}$ as all $\beta \in \mathbb{R}^d$, and their preference model as some ground-truth $\theta^* \in \mathcal{B}$ based on which they respond to all queries (potentially with noise in the behavioral model) **need to justify this somewhere – arguably various preference models are consistent with the naive decoder’s reasoning I think, let’s do this after we break the naive decoder in final form.**

4.1 Implications of linear frame restriction & extending utilities

Bailey: TODO

- What types of frames allowed
- Explain how generalizes the I matrix version of V and why this generalization is useful
- What it means to know V up front and why we could hope to do that for more general V

4.2 Reduction to existing methods

Write something here about the motivation for proving that we generalize the linear-over-attributes model. We want to show the following: fix the hypothesis class $\mathcal{B}$ and the optimal model $\beta^* \in \mathcal{B}$. We show that we can construct a V such that $\mathcal{B} \subseteq \Theta$ and, for the same V , there exists an instantiation of our model $\omega^*, \tau, \tau', (u_j | j \in J)$ such that β^* is the unique dominant rule. Constructing V is simple: v_j be the j -th standard basis vector (with a ϵ at the j -th position and 0s elsewhere). The learning goal of our model is to learn the frame weights ω_j on all the v_j ’s, which corresponds to the literature’s learning goal of learning the importance of each attribute.

Proposition 5 (For $1 < p < \infty$, any β^* can be made the unique maximizer). Fix $1 < p < \infty$ and let

$$\mathcal{B}_p = \{\beta \in \mathbb{R}^d : \|\beta\|_p = 1\}.$$

For any fixed $\beta^* \in \mathcal{B}_p$, there exists a vector $\omega^* \in \mathbb{R}^d$ with $\|\omega^*\|_1 = 1$ such that β^* is the *unique* maximizer of

$$\max_{\beta \in \mathcal{B}_p} \langle \omega^*, \beta \rangle.$$

In particular, one valid choice is

$$\tilde{\omega}_i^* = \text{sign}(\beta_i^*) |\beta_i^*|^{p-1}, \quad \omega^* = \frac{\tilde{\omega}^*}{\|\tilde{\omega}^*\|_1}.$$

PROOF. Let $\tilde{\omega}^*$ and ω^* be defined as in the statement, so $\|\omega^*\|_1 = 1$. Since scaling ω^* by any positive constant does not change the argmax, it suffices to show that β^* is the unique maximizer of $\langle \tilde{\omega}^*, \beta \rangle$ over $\|\beta\|_p = 1$.

Let q be the dual exponent, i.e. $\frac{1}{p} + \frac{1}{q} = 1$. By the standard duality lemma (Hölder duality), for any $\omega \neq 0$, the unique maximizer of $\max_{\|\beta\|_p=1} \langle \omega, \beta \rangle$ (when $1 < p < \infty$) is

$$\beta_i^{\text{opt}} = \frac{\text{sign}(\omega_i) |\omega_i|^{q-1}}{\|\omega\|_q^{q-1}}.$$

We plug in $\omega = \tilde{\omega}^*$. First note that for each i ,

$$\text{sign}(\tilde{\omega}_i^*) = \text{sign}(\beta_i^*), \quad |\tilde{\omega}_i^*| = |\beta_i^*|^{p-1}.$$

Therefore,

$$\text{sign}(\tilde{\omega}_i^*) |\tilde{\omega}_i^*|^{q-1} = \text{sign}(\beta_i^*) (|\beta_i^*|^{p-1})^{q-1}.$$

Now compute the exponent: since $q = \frac{p}{p-1}$, we have $q-1 = \frac{1}{p-1}$, hence

$$(|\beta_i^*|^{p-1})^{q-1} = |\beta_i^*|^{(p-1) \cdot \frac{1}{p-1}} = |\beta_i^*|.$$

So

$$\text{sign}(\tilde{\omega}_i^*) |\tilde{\omega}_i^*|^{q-1} = \text{sign}(\beta_i^*) |\beta_i^*| = \beta_i^*.$$

Thus the candidate optimizer produced by the duality lemma is

$$\beta^{\text{opt}} = \frac{\beta^*}{\|\tilde{\omega}^*\|_q^{q-1}}.$$

Since β^{opt} must satisfy $\|\beta^{\text{opt}}\|_p = 1$ and $\|\beta^*\|_p = 1$, it follows that the normalizing constant equals 1, so $\beta^{\text{opt}} = \beta^*$. Therefore β^* is the unique maximizer for $\tilde{\omega}^*$, and hence also the unique maximizer for $\omega^* = \tilde{\omega}^* / \|\tilde{\omega}^*\|_1$. $\square$

Proposition 6 (For $p = 1$, existence always holds, but uniqueness almost never holds). Let

$$\mathcal{B}_1 = \{\beta \in \mathbb{R}^d : \|\beta\|_1 = 1\}$$

be the full ℓ_1 sphere, and fix any $\beta^* \in \mathcal{B}_1$.

(Existence / weak optimality). There always exists $\omega^* \in \mathbb{R}^d$ with $\|\omega^*\|_1 = 1$ such that β^* is an optimizer of

$$\max_{\beta \in \mathcal{B}_1} \langle \omega^*, \beta \rangle.$$

One explicit choice is: let $S = \{i : \beta_i^* \neq 0\}$ and define

$$\tilde{\omega}_i^* = \begin{cases} \text{sign}(\beta_i^*) & \text{if } i \in S, \\ 0 & \text{if } i \notin S, \end{cases} \quad \omega^* = \frac{\tilde{\omega}^*}{\|\tilde{\omega}^*\|_1}.$$

Then $\|\omega^*\|_1 = 1$ and $\beta^* \in \arg \max_{\|\beta\|_1=1} \langle \omega^*, \beta \rangle$.

(Uniqueness characterization). There exists $\omega^* \in \mathbb{R}^d$ such that $\|\omega^*\|_1 = 1$ and β^* is the *unique* maximizer of $\max_{\|\beta\|_1=1} \langle \omega^*, \beta \rangle$ if and only if β^* is a signed standard basis vector, i.e.,

$$\beta^* \in \{\pm e_1, \dots, \pm e_d\}.$$

PROOF.

Existence. Let $S = \{i : \beta_i^* \neq 0\}$ and define $\tilde{\omega}^*$ and ω^* as in the statement. Then $\|\omega^*\|_1 = 1$ by construction. We show that β^* maximizes $\langle \omega^*, \beta \rangle$ over $\|\beta\|_1 = 1$.

First, for any β with $\|\beta\|_1 = 1$,

$$\langle \tilde{\omega}^*, \beta \rangle = \sum_{i \in S} \text{sign}(\beta_i^*) \beta_i \leq \sum_{i \in S} |\beta_i| \leq \sum_{i=1}^d |\beta_i| = \|\beta\|_1 = 1.$$

Next, evaluate at $\beta = \beta^*$:

$$\langle \tilde{\omega}^*, \beta^* \rangle = \sum_{i \in S} \text{sign}(\beta_i^*) \beta_i^* = \sum_{i \in S} |\beta_i^*| = \|\beta^*\|_1 = 1.$$

So β^* attains the upper bound, hence it is an optimizer for $\tilde{\omega}^*$. Scaling $\tilde{\omega}^*$ by a positive constant does not change the argmax set, so β^* is also an optimizer for $\omega^* = \tilde{\omega}^* / \|\tilde{\omega}^*\|_1$.

Uniqueness: “only if” direction. Assume β^* is the unique maximizer for some ω^* . Consider the ℓ_1 unit ball

$$B_1 := \{\beta \in \mathbb{R}^d : \|\beta\|_1 \leq 1\}.$$

Since $\beta \mapsto \langle \omega^*, \beta \rangle$ is linear, maximizing over the sphere $\{\|\beta\|_1 = 1\}$ is equivalent to maximizing over B_1 , and uniqueness over the sphere implies uniqueness over B_1 as well.

A linear functional achieves its maximum over a polytope at (at least) one extreme point. Moreover, if the maximizer is unique, it must be an extreme point of the feasible set. The extreme points of B_1 are exactly $\{\pm e_1, \dots, \pm e_d\}$. Therefore $\beta^* \in \{\pm e_1, \dots, \pm e_d\}$.

Uniqueness, “if” direction Now assume $\beta^* = s e_k$ for some $k \in \{1, \dots, d\}$ and $s \in \{+1, -1\}$. Fix any $\epsilon \in (0, 1)$ and define

$$\tilde{\omega}_i^* = \begin{cases} s & \text{if } i = k, \\ s\epsilon & \text{if } i \neq k, \end{cases} \quad \omega^* = \frac{\tilde{\omega}^*}{\|\tilde{\omega}^*\|_1}.$$

Then $\|\omega^*\|_1 = 1$. We claim that β^* is the unique maximizer of $\langle \omega^*, \beta \rangle$ over $\|\beta\|_1 = 1$. As scaling does not change argmax, it suffices to work with $\tilde{\omega}^*$.

For any β with $\|\beta\|_1 = 1$,

$$\langle \tilde{\omega}^*, \beta \rangle = s\beta_k + s\epsilon \sum_{i \neq k} \beta_i \leq |\beta_k| + \epsilon \sum_{i \neq k} |\beta_i| = |\beta_k| + \epsilon(1 - |\beta_k|),$$

since $\sum_i |\beta_i| = 1$. Define $t := |\beta_k| \in [0, 1]$. Then the bound becomes

$$\langle \tilde{\omega}^*, \beta \rangle \leq t + \epsilon(1 - t) = \epsilon + (1 - \epsilon)t.$$

Because $1 - \epsilon > 0$, this expression is strictly increasing in t , hence it is maximized only when $t = 1$, i.e. only when $|\beta_k| = 1$ and $\beta_i = 0$ for $i \neq k$. Under $\|\beta\|_1 = 1$, this forces $\beta = \pm e_k$. Finally, since the coefficient on coordinate k is s , we have

$$\langle \tilde{\omega}^*, s e_k \rangle = 1 \quad \text{and} \quad \langle \tilde{\omega}^*, -s e_k \rangle = -1,$$

so the unique maximizer is $\beta^* = s e_k$. □

This means that when $1 < p < \infty$, we can always find a set of frame weights in the instantiation where $v_j = e_j$ and there are d frames, 1 for each attribute, such that some given β^* is the UNIQUE dominant rule. If $p = 1$, we can always find a set of frame weights such that some given β^* is dominant, but it will not be the unique rule unless the given β^* is some basis vector.

Thus, not only do we capture the linear attribute model’s hypothesis class and preference model under one instantiation of V , we also generalize it: our model gives a mechanism for understanding a person’s ranking over all possible rules (since we have the frames, V , and the frame weights, we can calculate the preference relation between any two pairs of rules R, R' in the hypothesis class).

4.3 ω^* rule is non-dominated

Now, we show that at least for $p = 2$, the θ implied by ω^* is a non-dominated rule (and weakly dominant if $\tau, \tau' = 0$).

Bailey: I think if V is non-degenerate in some formalizable way, this is the unique dominant rule.

Lemma 1. Fix $V \in \mathbb{R}^{d \times m}$ with columns $\{v_j\}_{j=1}^m$. Let $\omega, \omega' \in \mathbb{R}^m$ satisfy $V\omega \neq 0$ and $V\omega' \neq 0$, and define

$$\theta := \frac{V\omega}{\|V\omega\|_2}, \quad \theta' := \frac{V\omega'}{\|V\omega'\|_2}.$$

Taking ω to be the “true” ω in our instance, we define the relative scores of the rule comparisons (these are the expressions implied by the utility functions):

$$s^+(\theta, \theta') := \sum_{j=1}^m \omega_j \langle v_j, \theta - \theta' \rangle \mathbf{1}(\langle v_j, \theta - \theta' \rangle > 0),$$

$$s^-(\theta, \theta') := \sum_{j=1}^m \omega_j \langle v_j, \theta' - \theta \rangle \mathbf{1}(\langle v_j, \theta' - \theta \rangle < 0).$$

Then

$$s^+(\theta, \theta') \geq s^-(\theta, \theta').$$

When $\tau, \tau' = 0$, this is equivalent to the event that $R_\theta \succ R_{\theta'}$.

PROOF. Let $a_j := \langle v_j, \theta - \theta' \rangle$. Note that $\langle v_j, \theta' - \theta \rangle = -a_j$. Hence

$$s^+(\theta, \theta') - s^-(\theta, \theta') = \sum_{j=1}^m \omega_j \left(a_j \mathbf{1}(a_j > 0) - (-a_j) \mathbf{1}(a_j < 0) \right).$$

Since $a_j \mathbf{1}(a_j > 0) + a_j \mathbf{1}(a_j < 0) = a_j$ (and the $a_j = 0$ case contributes 0),

$$s^+(\theta, \theta') - s^-(\theta, \theta') = \sum_{j=1}^m \omega_j a_j = \sum_{j=1}^m \omega_j \langle v_j, \theta - \theta' \rangle = \left\langle \sum_{j=1}^m \omega_j v_j, \theta - \theta' \right\rangle = \langle V\omega, \theta - \theta' \rangle.$$

Therefore

$$s^+ - s^- = \langle V\omega, \theta \rangle - \langle V\omega, \theta' \rangle.$$

Using $\theta = V\omega / \|V\omega\|_2$ gives

$$\langle V\omega, \theta \rangle = \left\langle V\omega, \frac{V\omega}{\|V\omega\|_2} \right\rangle = \frac{\|V\omega\|_2^2}{\|V\omega\|_2} = \|V\omega\|_2,$$

and using $\theta' = V\omega' / \|V\omega'\|_2$ gives

$$\langle V\omega, \theta' \rangle = \left\langle V\omega, \frac{V\omega'}{\|V\omega'\|_2} \right\rangle = \frac{\langle V\omega, V\omega' \rangle}{\|V\omega'\|_2}.$$

Hence

$$s^+ - s^- = \|V\omega\|_2 - \frac{\langle V\omega, V\omega' \rangle}{\|V\omega'\|_2}.$$

By Cauchy-Schwarz, $\langle V\omega, V\omega' \rangle \leq \|V\omega\|_2 \|V\omega'\|_2$, so

$$\frac{\langle V\omega, V\omega' \rangle}{\|V\omega'\|_2} \leq \|V\omega\|_2,$$

and thus $s^+ - s^- \geq 0$, i.e. $s^+(\theta, \theta') \geq s^-(\theta, \theta')$. □

Bailey: I think there's a generalization beyond the dot product where this works for arbitrary norms (because Cauchy-Schwarz exists for generic norms), but I think let's stick with the 2-norm.

Bailey: Should figure out condition for which θ^* is the unique dominant rule / when it is dominant rather than conflicted.

4.4 Lipschitzness between the two spaces

That for two very similar omegas, the rules they imply will have very low s^+ scores. This will allow us to claim that if we reduce the diameter of Ω sufficiently, we have also reduced the rule space to a “small” set as it relates to preference error.

Definition 4.4 (V -only nondegeneracy and size constants). Fix $V \in \mathbb{R}^{d \times m}$ with columns $\{v_j\}_{j=1}^m$ and define the simplex in which all ω 's live:

$$\Delta_m := \{\omega \in \mathbb{R}^m : \omega \geq 0, \mathbf{1}^\top \omega = 1\}.$$

Define

$$\alpha_V := \min_{\omega \in \Delta_m} \|V\omega\|_2 = \text{dist}(0, \text{conv}\{v_1, \dots, v_m\}), \quad L := \max_{j \in [m]} \|v_j\|_2.$$

We assume $\alpha_V > 0$ (equivalently $0 \notin \text{conv}\{v_1, \dots, v_m\}$).

Lemma 2 (Lipschitz control and smallness of $s^\pm$ on the simplex). Fix $V \in \mathbb{R}^{d \times m}$ with columns $\{v_j\}_{j=1}^m$ and let the omegas be norm-1 normalized, so they live in

$$\Delta_m := \{\omega \in \mathbb{R}^m : \omega \geq 0, \mathbf{1}^\top \omega = 1\}.$$

Define the V -only constants

$$\alpha_V := \min_{\omega \in \Delta_m} \|V\omega\|_2 = \text{dist}(0, \text{conv}\{v_1, \dots, v_m\}), \quad L := \max_{j \in [m]} \|v_j\|_2,$$

and assume $\alpha_V > 0$ (equivalently $0 \notin \text{conv}\{v_1, \dots, v_m\}$). For any $\omega \in \Delta_m$ with $V\omega \neq 0$, define the normalized image

$$\theta(\omega) := \frac{V\omega}{\|V\omega\|_2}.$$

Let $\omega', \omega'' \in \Delta_m$ and set $\theta' := \theta(\omega')$ and $\theta'' := \theta(\omega'')$.

For any $\omega \in \Delta_m$, define

$$s^+(\theta', \theta''; \omega) := \sum_{j=1}^m \omega_j \langle v_j, \theta' - \theta'' \rangle \mathbf{1}(\langle v_j, \theta' - \theta'' \rangle > 0),$$

$$s^-(\theta', \theta''; \omega) := \sum_{j=1}^m \omega_j \langle v_j, \theta'' - \theta' \rangle \mathbf{1}(\langle v_j, \theta'' - \theta' \rangle < 0).$$

Then:

(i) (Normalization is Lipschitz on Δ_m in ℓ_1)

$$\|\theta' - \theta''\|_2 \leq \frac{2L}{\alpha_V} \|\omega' - \omega''\|_1.$$

(ii) ($s^\pm$ are controlled by $\|\theta' - \theta''\|_2$)

$$s^+(\theta', \theta''; \omega) + s^-(\theta', \theta''; \omega) \leq L \|\theta' - \theta''\|_2$$

(iii) (Combined bound: closeness of ω implies small $s^\pm$)

$$s^+(\theta', \theta''; \omega) \leq \frac{2L^2}{\alpha_V} \|\omega' - \omega''\|_1, \quad s^-(\theta', \theta''; \omega) \leq \frac{2L^2}{\alpha_V} \|\omega' - \omega''\|_1.$$

PROOF. Because $\alpha_V = \min_{\omega \in \Delta_m} \|V\omega\|_2 > 0$, we have $\|V\omega'\|_2 \geq \alpha_V$ and $\|V\omega''\|_2 \geq \alpha_V$. The standard normalization inequality for nonzero $x, y \in \mathbb{R}^d$ states

$$\left\| \frac{x}{\|x\|_2} - \frac{y}{\|y\|_2} \right\|_2 \leq \frac{2\|x - y\|_2}{\min\{\|x\|_2, \|y\|_2\}}.$$

Apply this with $x = V\omega'$ and $y = V\omega''$ to get

$$\|\theta' - \theta''\|_2 \leq \frac{2\|V(\omega' - \omega'')\|_2}{\alpha_V}.$$

Now write $\delta := \omega' - \omega''$ and use the triangle inequality:

$$\|V\delta\|_2 = \left\| \sum_{j=1}^m \delta_j v_j \right\|_2 \leq \sum_{j=1}^m |\delta_j| \|v_j\|_2 \leq \left(\max_{j \in [m]} \|v_j\|_2 \right) \sum_{j=1}^m |\delta_j| = L \|\delta\|_1.$$

Combining yields (i).

For (ii), let $\Delta := \theta' - \theta''$ and $a_j := \langle v_j, \Delta \rangle$. Then

$$s^+(\theta', \theta''; \omega) + s^-(\theta', \theta''; \omega) = \sum_{j=1}^m \omega_j |a_j|.$$

By Cauchy-Schwarz, $|a_j| \leq \|v_j\|_2 \|\Delta\|_2 \leq L \|\Delta\|_2$, hence using $\sum_{j=1}^m \omega_j = 1$ (since $\omega \in \Delta_m$) gives

$$s^+(\theta', \theta''; \omega) + s^-(\theta', \theta''; \omega) \leq \sum_{j=1}^m \omega_j L \|\Delta\|_2 = L \|\Delta\|_2.$$

Since $s^+, s^- \geq 0$, the bounds in (ii) follow.

Finally, (iii) follows by combining (ii) with (i). □

Bailey: Why is rule set non-convex if lipschitz

Okay, now we have some intuition that if we get close on ω , we can get close on θ . We know that under non-degenerate V , it can be true that there is a unique dominant rule, so we need to formulate our learning goal to handle this case.

Define $\theta^* := V\omega^*/\|V\omega^*\|_2$ to be the optimal rule (even if it's non-uniquely dominant, we will use it as our target).

Bailey: This uses general norms but that is not for a reason – we should be setting $p = 2$

Bailey: Definitely need to show that ϵ -PO rule always exists (or even PO) – just gives you existence but should state

4.5 Small Ω implies near pareto optimality

Recall our space of thetas:

$$\Theta := \{\theta \in \mathbb{R}^d : \exists \omega \in \Omega : \theta = \frac{V\omega}{\|V\omega\|_p} | \omega \in \Omega\}$$

Now, we have some space of rules as our hypothesis class,

$$\mathcal{H} := \{R_\theta | \theta \in \Theta\}$$

Our learning goal is to learn a set set of ϵ -pareto optimal rules (or just one, if a set is not possible). I think by the above, we pretty much just need to shrink Ω to have small diameter (stretched in the directions of V , I believe), and we will be able to guarantee this.

Setup. For any $\Omega_t \subseteq \Omega$, define the induced parameter set

$$\Theta_t := \left\{ \theta(\omega) := \frac{V\omega}{\|V\omega\|_p} : \omega \in \Omega_t \right\}, \quad \mathcal{H}_t := \{R_\theta : \theta \in \Theta_t\} \subseteq \mathcal{H}.$$

Let the *diameter* of Ω_t be

$$\text{diam}(\Omega_t) := \sup_{\omega, \omega' \in \Omega_t} \|\omega - \omega'\|_1.$$

(Any norm can be used, using $\|\cdot\|_1$ for correspondence to lemma above.)

Define the column-scale parameter

$$s_V := \max_{j \in [m]} \|v_j\|_p,$$

where v_k denotes the k th column of V . Also define the *nondegeneracy margin* on Ω_t :

$$\alpha_t := \inf_{\omega \in \Omega_t} \|V\omega\|_p > 0.$$

Utility-Lipschitzness in θ : Fix a frame $j \in [m]$. By definition, $u_j(R_\theta) = \langle v_j, \theta \rangle$. Hence for any $\theta, \theta' \in \Theta$,

$$|u_j(R_\theta) - u_j(R_{\theta'})| = |\langle v_j, \theta - \theta' \rangle| \leq \|v_j\|_q \|\theta - \theta'\|_p, \quad (7)$$

where q is the Hölder conjugate of p (i.e. $\frac{1}{p} + \frac{1}{q} = 1$, with the conventions $p = 1 \Rightarrow q = \infty$ and $p = \infty \Rightarrow q = 1$). Thus $u_j(\cdot)$ is L_j -Lipschitz in θ with

$$L_j := \|v_j\|_q.$$

To get a single constant valid for all $j \in [m]$, define

$$L := \max_{j \in [m]} \|v_j\|_q.$$

Then for all $j \in [m]$ and all $\theta, \theta' \in \Theta$,

$$|u_j(R_\theta) - u_j(R_{\theta'})| \leq L_j \|\theta - \theta'\|_p \leq L \|\theta - \theta'\|_p$$

Lemma 3 (Normalization map is Lipschitz on Ω_t). For all $\omega, \omega' \in \Omega_t$,

$$\left\| \frac{V\omega}{\|V\omega\|_p} - \frac{V\omega'}{\|V\omega'\|_p} \right\|_p \leq \frac{2\|V(\omega - \omega')\|_p}{\alpha_t} \leq \frac{2s_V}{\alpha_t} \|\omega - \omega'\|_1.$$

Consequently,

$$\text{diam}(\Theta_t) := \sup_{\theta, \theta' \in \Theta_t} \|\theta - \theta'\|_p \leq \frac{2s_V}{\alpha_t} \sup_{\omega, \omega' \in \Omega_t} \|\omega - \omega'\|_1 = \frac{2s_V}{\alpha_t} \text{diam}(\Omega_t).$$

THEOREM 3 (SHRINKING Ω_t IMPLIES ε_t -PARETO-UNDOMINANCE IN $\mathcal{H}_t$). Assume $\alpha_t > 0$. Define

$$\varepsilon_t := s_V^{(q)} \cdot \text{diam}_p(\Theta_t) \leq s_V^{(q)} \cdot \frac{2s_V^{(p)}}{\alpha_t} \text{diam}_1(\Omega_t). \quad (8)$$

Then every $R_\theta \in \mathcal{H}_t$ is ε_t -Pareto-optimal: there does not exist $R_{\theta'} \in \mathcal{H}_t$ such that

$$u_j(R_{\theta'}) \geq u_j(R_\theta) + \varepsilon_t \quad \forall j \in [m].$$

Equivalently, $\mathcal{H}_t \subseteq_{\varepsilon_t} (\mathcal{H}_t)$.

PROOF. Fix any $\theta, \theta' \in \Theta_t$ and any frame $j \in [m]$. By equation (7),

$$u_j(R_{\theta'}) - u_j(R_\theta) \leq |u_j(R_{\theta'}) - u_j(R_\theta)| \leq s_V^{(q)} \|\theta' - \theta\|_p.$$

Taking the supremum over $\theta, \theta' \in \Theta_t$ yields

$$u_j(R_{\theta'}) - u_j(R_\theta) \leq s_V^{(q)} \text{diam}_p(\Theta_t) = \varepsilon_t \quad \forall j \in [m], \forall \theta, \theta' \in \Theta_t.$$

Therefore, for any fixed $\theta \in \Theta_t$, there cannot exist $\theta' \in \Theta_t$ satisfying $u_j(R_{\theta'}) \geq u_j(R_\theta) + \varepsilon_t$ for all j : such a θ' would contradict the inequality above for each j . Hence every $R_\theta \in \mathcal{H}_t$ is ε_t -Pareto-undominated in $\mathcal{H}_t$.

Finally, the bound $\varepsilon_t \leq s_V^{(q)} \cdot \frac{2s_V^{(p)}}{\alpha_t} \text{diam}_1(\Omega_t)$ follows by substituting equation (??) into equation (8). $\square$

Put into a more direct statement form:

COROLLARY 4. Assume $\alpha_t := \inf_{\omega \in \Omega_t} \|V\omega\|_p > 0$. If $\text{diam}_1(\Omega_t) \leq \varepsilon$, then every $R_\theta \in \mathcal{H}_t$ is $f(\varepsilon)$ -Pareto-optimal in $\mathcal{H}_t$ under the uniform-margin notion, where

$$f(\varepsilon) := \frac{2s_V^{(p)} s_V^{(q)}}{\alpha_t} \varepsilon.$$

4.6 Geometry of objects

Now, our goal is to devise a learning algorithm for shrinking the omega space quickly, in a way that shrinks the rule space as fast as possible.

Geometry of Ω_t . Let $\Omega_0 := \Omega \subseteq \mathbb{R}^m$ be the initial feasible set of frame-weights, assumed convex (e.g. a simplex or an intersection of affine halfspaces). At time $t \geq 1$, the learner poses a comparison query $q_t = (X_t, Y_t, Y'_t)$ and observes a response $a_t \in \{\succ, \prec, \sim, \bowtie\}$.

Under the finite-edits extension, the within-frame replacement gap at $(X_t; Y_t, Y'_t)$ is the known constant

$$\Delta_{j,t} := u_j(R[X_t \mapsto Y_t]) - u_j(R[X_t \mapsto Y'_t]) = \lambda(X_t) \langle v_j, \phi(X_t, Y_t) - \phi(X_t, Y'_t) \rangle.$$

Define the aggregate signed gap and total intensity

$$\Delta_t(\omega) := \sum_{j=1}^m \omega_j \Delta_{j,t}, \quad r_t(\omega) := \sum_{j=1}^m \omega_j |\Delta_{j,t}|.$$

(Note that $\Delta_t(\omega) = s_t^+(\omega) - s_t^-(\omega)$ and $r_t(\omega) = s_t^+(\omega) + s_t^-(\omega)$.) Then the (τ, τ') -threshold rule can be written as linear constraints in ω :

- If a_t is *strict preference* $Y_t \succ Y'_t$, then

$$r_t(\omega) \geq \tau \quad \text{and} \quad \Delta_t(\omega) \geq \tau' r_t(\omega).$$

Equivalently,

$$\sum_{j=1}^m \omega_j |\Delta_{j,t}| \geq \tau, \quad \sum_{j=1}^m \omega_j (\Delta_{j,t} - \tau' |\Delta_{j,t}|) \geq 0.$$

- If a_t is *strict reverse preference* $Y_t \prec Y'_t$, then

$$r_t(\omega) \geq \tau \quad \text{and} \quad \Delta_t(\omega) \leq -\tau' r_t(\omega).$$

Equivalently,

$$\sum_{j=1}^m \omega_j |\Delta_{j,t}| \geq \tau, \quad \sum_{j=1}^m \omega_j (\Delta_{j,t} + \tau' |\Delta_{j,t}|) \leq 0.$$

- If a_t is *indifference* $Y_t \sim Y'_t$, then

$$r_t(\omega) < \tau,$$

(which can be implemented as $r_t(\omega) \leq \tau$ or $r_t(\omega) \leq \tau - \eta$ for a small $\eta > 0$).

- If a_t is *incomparability* $Y_t \bowtie Y'_t$, then

$$r_t(\omega) \geq \tau \quad \text{and} \quad |\Delta_t(\omega)| < \tau' r_t(\omega).$$

Equivalently,

$$\sum_{j=1}^m \omega_j |\Delta_{j,t}| \geq \tau, \quad \sum_{j=1}^m \omega_j (\Delta_{j,t} - \tau' |\Delta_{j,t}|) < 0, \quad \sum_{j=1}^m \omega_j (\Delta_{j,t} + \tau' |\Delta_{j,t}|) > 0.$$

We update by intersection:

$$\Omega_t := \Omega_{t-1} \cap C(q_t, a_t), \quad \Omega_T = \Omega \cap \bigcap_{t=1}^T C(q_t, a_t).$$

Geometric interpretation. For fixed (X_t, Y_t, Y'_t) the coefficients $\Delta_{j,t}$ are constants, so both $\Delta_t(\omega)$ and $r_t(\omega)$ are linear functions of ω . Hence each observation a_t contributes one or more halfspace constraints (or open halfspaces for strict inequalities). Thus Ω_t remains convex for all t (a polyhedron, and a polytope if Ω is bounded), and each informative query carves away a slab/halfspace of weight vectors whose implied $(r_t(\omega), \Delta_t(\omega))$ would have produced a different response.

Geometry of Θ_t While Ω_t is an affine polytope, Θ_t is not exactly convex. Let $Z_t := V\Omega_t \subseteq \mathbb{R}^d$. Since V is linear and Ω_t is a (polyhedral) convex set, Z_t is itself a convex polytope/polyhedron. For $p = 2$, the induced parameter set

$$\Theta_t = \left\{ \frac{V\omega}{\|V\omega\|_2} : \omega \in \Omega_t \right\} = \left\{ \frac{z}{\|z\|_2} : z \in Z_t, z \neq 0 \right\} \subseteq \mathbb{S}^{d-1}$$

is the *radial projection* of Z_t onto the unit sphere. Thus Θ_t can be interpreted as the set of *directions* (unit vectors) attained by points in Z_t , equivalently the set of rays from the origin that intersect Z_t . Although Θ_t is not convex in the ambient Euclidean sense (convex combinations generally leave the sphere), it retains substantial structure: it is the intersection of the sphere with the cone generated by Z_t ,

$$\Theta_t = \mathbb{S}^{d-1} \cap \text{cone}(Z_t), \quad \text{cone}(Z_t) := \{\lambda z : \lambda \geq 0, z \in Z_t\},$$

and its boundary on the sphere is the radial image of faces of the polytope Z_t , yielding a “spherical polytope”-like region composed of finitely many spherical facets. Moreover, when Z_t is contained in an open halfspace (equivalently, Θ_t lies in a hemisphere, as is typical under the nondegeneracy condition $\alpha_t > 0$), Θ_t is often geodesically convex on $\mathbb{S}^{d-1}$: for any two points in Θ_t , the shorter great-circle arc between them remains in Θ_t .

4.7 Learning Algorithms

Learning algorithm (perfect information). First, let’s suppose that we know τ, τ' and all frames and all corresponding features, so all we don’t know is the true ω^* .

It is inefficient to naively try to shrink Ω_t because you might be shrinking it on degenerate dimensions of V , i.e., in directions where the choice of ω doesn’t affect or barely affects the rule. We therefore want to shrink Z_t , and because our actual set Θ_t is Z_t down-projected onto the sphere, there is a close relationship between shrinking Z_t and Θ_t .

Lemma 4 (From shrinking $Z_t = V\Omega_t$ to shrinking Θ_t). Let $p = 2$ and define $Z_t := V\Omega_t \subseteq \mathbb{R}^d$ and

$$\Theta_t = \left\{ \frac{z}{\|z\|_2} : z \in Z_t, z \neq 0 \right\} \subseteq \mathbb{S}^{d-1}.$$

Assume the nondegeneracy margin

$$\alpha_t := \inf_{z \in Z_t} \|z\|_2 > 0.$$

Then

$$\text{diam}_2(\Theta_t) := \sup_{\theta, \theta' \in \Theta_t} \|\theta - \theta'\|_2 \leq \frac{2}{\alpha_t} \text{diam}_2(Z_t), \quad \text{diam}_2(Z_t) := \sup_{z, z' \in Z_t} \|z - z'\|_2.$$

PROOF. Fix any nonzero $z, z' \in Z_t$ and let $\theta = z/\|z\|_2$, $\theta' = z'/\|z'\|_2$. Write

$$\theta - \theta' = \frac{z - z'}{\|z\|_2} + z' \left(\frac{1}{\|z\|_2} - \frac{1}{\|z'\|_2} \right).$$

Taking norms and using $|\|z\|_2 - \|z'\|_2| \leq \|z - z'\|_2$ gives

$$\begin{aligned} \|\theta - \theta'\|_2 &\leq \frac{\|z - z'\|_2}{\|z\|_2} + \|z'\|_2 \cdot \frac{|\|z\|_2 - \|z'\|_2|}{\|z\|_2 \|z'\|_2} \\ &\leq \frac{\|z - z'\|_2}{\|z\|_2} + \frac{\|z - z'\|_2}{\|z\|_2} = \frac{2\|z - z'\|_2}{\|z\|_2}. \end{aligned}$$

By symmetry the same bound holds with $\|z'\|_2$ in the denominator, hence

$$\|\theta - \theta'\|_2 \leq \frac{2\|z - z'\|_2}{\min\{\|z\|_2, \|z'\|_2\}} \leq \frac{2\|z - z'\|_2}{\alpha_t}.$$

Taking the supremum over $z, z' \in Z_t$ yields $\text{diam}_2(\Theta_t) \leq \frac{2}{\alpha_t} \text{diam}_2(Z_t)$. □

COROLLARY 5 (DETERMINISTIC $f(\delta)$ -PARETO GUARANTEE VIA Z_t). Assume $u_j(R_\theta) = \langle v_j, \theta \rangle$ and let $L := \max_{j \in [m]} \|v_j\|_2$. If $\text{diam}_2(Z_t) \leq \delta$ and $\alpha_t > 0$, then every rule in $\mathcal{H}_t = \{R_\theta : \theta \in \Theta_t\}$ is ε_t -Pareto-undominated in $\mathcal{H}_t$ under the uniform-margin notion, with

$$\varepsilon_t \leq L \text{diam}_2(\Theta_t) \leq \frac{2L}{\alpha_t} \delta.$$

Now, we need to define a learning algorithm for finding the next query. Fortunately, we are shrinking an affine polytope.

Bailey: Michelle – see if you can find a standard method for this, especially for generic but known τ, τ' .

Learning algorithm (unknown frames) Should work exactly as I wrote in main.

- Some model of how people reveal frames in initial queries – we get most of the important ones
- **How do we find out V ?**
- After queries, we can ask people which frames were active

4.8 Breaking the naive decoder

Bailey: Break in simplest linear case possible (reduction case ideally)

Assume $V = I$. The hypothesis class is $\theta = \frac{V\omega}{\|V\omega\|_p}$. We do not ask for frames, we ONLY ask pairwise comparisons and we don't get what frames are activated per pairwise comparison.. Problems: we don't know the frames in general, and we don't know what frames each query activates.

First, we have to define the violation function. The only input into the violation function we have is a pairwise comparison query and the agent's choice.

To recap, here's what is true based on one observed pairwise comparison for our model, except under linear rules:

Definition 4.5 (Global Outcome Scores). Fix a query $q = (X, Y, Y')$. Let

$$U_j(Y) = \sum_R u_j(R[X \rightarrow Y]) = \sum_{\theta \in \mathcal{H}} u_j(R_\theta) + \lambda(X) \left(\langle v_j, \phi(X, Y) \rangle - \langle v_j, \phi(X, R_\theta(X)) \rangle \right)$$

$$U_j(Y') = \sum_R u_j(R[X \rightarrow Y']) = \sum_{\theta \in \mathcal{H}} u_j(R_\theta) + \lambda(X) \left(\langle v_j, \phi(X, Y') \rangle - \langle v_j, \phi(X, R_\theta(X)) \rangle \right)$$

Then define the scores of different answers as

$$s^+(Y, Y') := \sum_{j: Y \succ_j Y'} \omega_j (U_j(Y) - U_j(Y')) = \sum_{j: Y \succ_j Y'} \omega_j \lambda(X) \langle v_j, (\phi(X, Y) - \phi(X, Y')) \rangle$$

$$s^-(Y, Y') := \sum_{j: Y \prec_j Y'} \omega_j (U_j(Y') - U_j(Y)) = \sum_{j: Y \prec_j Y'} \omega_j \lambda(X) \langle v_j, (\phi(X, Y') - \phi(X, Y)) \rangle$$

Definition 4.6 (Query-Model Outcome Preferences). We again define the quantities we will threshold:

$$r = s^+ + s^-, \quad \theta = \frac{s^+ - s^-}{s^+ + s^-}$$

Now, we can define analogous conditions to global preferences over rules, now over Y, Y' with respect to a single query X :

- $Y \succ_{\mathcal{M}, X} Y' \iff r \geq \tau \text{ and } \theta \geq \tau'$
- $Y' \succ_{\mathcal{M}, X} Y \iff r \geq \tau \text{ and } \theta \leq -\tau'$
- $Y \bowtie_{\mathcal{M}, X} Y' \iff r \geq \tau \text{ and } \theta \in (-\tau', \tau')$